antimicrobial agents in tubes with standard amount of bacteria. Turbidity (cloudiness) after incubation indicates bacterial growth and lack of turbidity indicates that the growth of bacteria is inhibited.

Etest

This is another test to determine the minimum inhibitory concentration where a plastic strip containing a gradient of the antimicrobial agent is used (Figure 3.4). An elliptical zone of inhibitory concentration can be noted with the help of a scale printed on the strip.

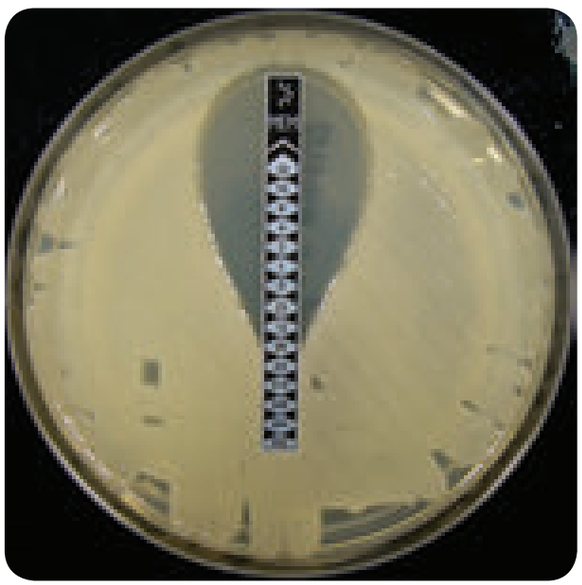


Figure 3.4: E – test

The Minimal Bactericidal Concentration (MBC) Test

MBC test is similar to MIC, the minimal bactericidal concentration test is used to determine the amount of antimicrobial agent required to rather **kill** the pathogen inhibit its growth. In MBC test, samples taken from MIC tubes are transferred to drug free plates. Bacterial growth in these subcultures indicates that some bacterial cells have survived antimicrobial drug. The lowest concentration of drug for which no growth occurs is the minimum bactericidal concentration.

The tube dilution method is considered accurate for determining susceptibility of a pathogen to precise quantities antimicrobial agent. However, the method is time consuming, expensive, and not practical for use in most clinical laboratories for routine susceptibility testing.

Infobits

What is CRE?

CRE, which stands for carbapenem resistant Enterobacteriaceae, is the most fearsome family of germs because it is resistant even to last-resort antiboitcs.

3.8 Drugs Resistance Mechanisms

Some microbes respond predictably to certain drugs making selection of treatment easy. Other microbes may vary in their responses, and laboratory tests are usually required to ensure that the selected therapy is appropriate. Chemotherapeutic effectiveness depends upon the sensitivity of the pathogen to the agent. Antibiotic resistance, however, may develop in microbes within the population. In fact, the history of chemotherapy has been closely paralleled by the history of drug resistance.

None of the therapeutic drugs (antibiotic) inhibits all microbial pathogens and some microbial pathogens possess natural ability to resist to certain antibiotics. Bacteria become drugs resistant using several different resistance mechanisms. A particular type of resistance mechanism is not confined to a single class of drugs. Two bacteria may employ different resistance mechanisms to counter the same antibiotic.

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